NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.3b AI-hybrid synthetic typhoons to coastal design wind speeds

Group AI Weather Models

Status Closed (negative result)

Period 2026-10-14 19:18 — 2026-10-24 22:03

Stage conclusion

Project report

B2 AI-hybrid synthetic tropical cyclones for coastal design wind speeds

Objective

Use an AI/ML-assisted synthetic tropical cyclone catalog (RAFT: statistical tracks + MLP intensity) coupled with the parameterized wind field that passed Gate (a) in B1+B4 (CLIMADA 6.1.0 H08 + roughness) to compute 50- and 700-year return-period wind speeds at US Gulf and Atlantic coastal stations; compare these against the traditional statistical catalog STORM run through the same wind field chain and against ASCE 7-22 design wind speeds. Determine whether "AI synthetic tropical cyclones → design wind speeds" is worth continuing (the wind field model will be replaced once the PI's model is ready). Task brief.

Criteria (frozen 10-14 19:25 after both EVA and Design Loads tracks replied; no post-hoc relaxation permitted)

Unified convention: Station height 10 m, open terrain (z0 = 0.03 m, corresponding to ASCE Exposure C), 3 s gust, denoted V50 and V700 (MRI 50 and 700 years). Basis: Vickery et al. 2010 give 300/700/1,700-year maps as peak gusts at 10 m in open terrain (z0=0.03 m), where Category II is 700 years (card, PDF pp. 1, 5–8 / journal pp. 613, 617–620; that paper does not explicitly state "3 s"); "3 s" is taken from ASCE 7-22 Chapter 26, where the basic wind speed V in ASCE 7-22 is the nominal 3 s gust at 10 m (33 ft) in Exposure C (close reading, Ch. 26; exact page number mapping Risk Category II to 700 years pending verification against the original text by ideas). All stations are uniformly computed assuming open terrain, without directional station-exposure corrections (the ASCE maps themselves assume uniform Exposure C).

- **Gate 0 (wind field chain qualification)**: Across stations, the median bias of V700 from the STORM chain relative to ASCE 7-22 must be within ±20%. If it fails → record the conclusion as "wind field / conversion chain unqualified" and do not evaluate AI (same logic as Gate (a) in B1+B4).
- **Gate 1 (worth continuing)**: Across stations, the median absolute bias of V700 from the RAFT chain relative to ASCE 7-22 must be ≤ 15%, **and** ≤ 1.1 × the median absolute bias of the STORM chain. If both are satisfied → worth continuing; otherwise put it on hold and clearly document where the discrepancy lies (catalog intensity/frequency, wind field model, or conversion convention).
- **Report-only, non-gating**: Three-way V50 comparison; breakdown by coastal segment (Texas–Louisiana / Mississippi–Florida Panhandle / Florida Peninsula & Keys / Georgia–North Carolina); sampling uncertainty of return periods (annual-resampling bootstrap); sensitivity to averaging-time / gust conversions and the STORM 10 min→1 min conversion; if a public SPIN or Bose 2023 catalog is found, include it as a third chain for reporting only.
- **Decision rule if ASCE values are unavailable (pre-specified at freeze)**: ASCE 7-22 station values are retrieved strictly from public sources (ATC Hazards by Location, etc., without using the PI's credentials; no reading off code PDF maps to make up numbers). If ≥12 out of 19 stations are obtained → compute and judge Gate 0 and Gate 1 on the obtained stations as usual. If <12 stations are obtained → do not judge Gate 0 / Gate 1; downgrade to a **relative comparison**: if the across-station median absolute difference in V700 of RAFT relative to STORM is ≤15%, and the median difference in each of the four coastal segments is within ±25%, record "RAFT is consistent with the traditional catalog and may continue (untested against code)"; otherwise record "RAFT is inconsistent with the traditional catalog" and specify where the discrepancy lies. Conclusions from a relative comparison must not be stated as equivalent to design wind speeds.
- **Design Loads track materials are read-only** (10-14 reply from the design loads lead): may cite the ASCE 7-22 wind interface close reading (needs_review), its page mapping (PDF page = print page + 61), and the code text itself. Do not modify their calculators; do not cite or use for endorsement the numerical artifacts named "B2/C0" on their track (those belong to their own line, not this node).
- **Stratified reporting (added by parent project on 10-14, report-only, non-gating)**: H08 is known to overestimate across broad outer regions by ~+2.5 m/s (Node H) and underestimate in the inner core <1 RMW. For each station's V50 and V700, report the breakdown of exceedance contributions by storm track distance to the station in bins of <1, 1–3, and >3 × RMW (classified by distance/RMW at the time of

peak station wind among storms exceeding that return-period wind speed); additionally, stratify and report station biases by distance to the coast (<10, 10–50, >50 km).
- Constants, stations, and methods must be written into a timestamped record before computing any station wind speeds; no changes afterward; any required modification must be logged as v2 while preserving v1.

Stations (frozen 10-14 19:25, 19 ASOS airport stations from Texas to North Carolina)

KBRO Brownsville, KCRP Corpus Christi, KGLS Galveston, KLCH Lake Charles, KMSY New Orleans, KGPT Gulfport, KMOB Mobile, KPNS Pensacola, KAAF Apalachicola, KTPA Tampa, KRSW Fort Myers, KEYW Key West, KMIA Miami, KPBI West Palm Beach, KMLB Melbourne, KJAX Jacksonville, KCHS Charleston, KILM Wilmington, KHSE Cape Hatteras. Only station coordinates are used (IEM station table); ASOS wind observations are not needed. If an ASCE value cannot be retrieved for a station, record it as missing and do not substitute the station.

Current status

Stop category: Closed (negative result), upheld by the 10-21 six-reviewer stop-category review (5 valid opinions: 3 in favor, 2 opposed; verdict). Negated configuration (strictly limited to this one): Public RAFT North Atlantic catalog (Zenodo 10392725, 40,000 tracks, including its Pc and Rmax empirically derived from vmax) + CLIMADA 6.1.0 H08 (which does not read catalog vmax) + POCI 1010 hPa, 300 km cutoff, 15 min interpolation + 1.07 conversion factor + HURDAT2 observational proxy annual rate of 12.075 storms/year (10–20 sensitivity yields the same conclusion), across 19 frozen coastal stations, comparing V700 against ASCE 7-22 (TC-only against all-hazard values, shared across both chains). **No extrapolation**: This does not negate RAFT's MLP intensity model itself, wind field models that directly use vmax (such as H10, untested), or the overall "AI synthetic tropical cyclones → design wind speed" pathway. **10-14 22:30: put on hold** (6/6 in favor; verdict). - **Results** (primary version v6, RAFT annual rate set to HURDAT2 observational proxy 12.075 storms/year): Gate 0 passed (STORM V700 vs. ASCE 7-22 median −8.87%, 19/19 stations); Gate 1 failed (RAFT median absolute bias 30.95%, >15% and >1.1×9.21%). RAFT is biased low across all four coastal segments; V50 (18 stations, KEYW has no code value) has a median bias of −31.2% (v6, supported by ~66 storms per station), showing negative bias across the entire distribution; the upper bounds of the 90% intervals for RAFT V700 across all 19 stations fall below ASCE, with the closest station (KJAX) still 15.55% low (v6 recalculation; the 10.9% cited in the earlier verdict was from the 14.5-storm/yr run). Varying the annual rate across 10–20, POCI by ±5 hPa, or the conversion factor across 1.03–1.12 does not alter the verdict. - **Why it failed**: The shared wind field chain is controlled by STORM passing Gate 0, and units/inputs/plotting were verified against CLIMADA 6.1.0 source code by 4 reviewers, finding no implementation error capable of explaining a 30% discrepancy. Because H08 uses only pressure deficit and does not read catalog vmax, this chain actually tests **the empirical Pc/Rmax formulas supplied by RAFT**, not its AI intensity model per se. Whether the low bias stems from the empirical Pc–vmax relation (intercept 1030 hPa paired with POCI 1010), downward-biased Rmax regression, nearshore

decay, or shared assumptions was **not quantitatively separated** in this round; the authors' self-reported biases (Rmax ~19% low, landfall 99th percentile ~8 kt low, RAFT card, PDF pp. 6–7) serve only as directional clues. - **What is missing**: A wind field model that directly ingests vmax (or a self-consistent Pc–vmax relationship), plus a distributional comparison of RAFT Pc/Rmax/nearshore intensity against HURDAT2/EBTRK. - **Conditions for restart**: Prerequisite work: either the PI's wind field model becomes available, or a distributional diagnostic of RAFT Pc/Rmax/nearshore intensity against HURDAT2/EBTRK is completed first (initial hypothesis: pairing the linear Pc formula with purely pressure-driven H08 loses ~10–15% at the strong-storm end, estimated by reviewer E, not yet recomputed). **Trigger**: write a new pre-registration based on those findings and re-run under the same frozen thresholds; restart succeeds only if Gate 0 still passes and Gate 1 passes, without tuning any parameters against this round's ASCE errors. If a new public AI synthetic catalog emerges that natively includes Pc/Rmax (which SPIN currently lacks), it may also be tested under the same protocol. - **10-15 00:10 B2 wrap-up (status remains On hold); wording narrowed on 10-21 per the PI's approval**: Under this configuration (RAFT catalog + CLIMADA H08 + this conversion chain, 19 stations, V700 vs. ASCE 7-22), the input discrepancy **correlated** with the negative bias is fewer near-station passages within 100 km and weaker intensity during passage (19-station mean TS+ 0.168 vs. observed 0.321 storms/station/year, Cat3+ 0.0104 vs. 0.0374; Cat3+ share 6.2% vs. 11.6%, STORM 16.0%; most pronounced in the Gulf of Mexico). **No separation experiment was performed**: within passage frequency, track geometry cannot be separated from the overall occurrence rate, nor can wind-speed bias percentages be back-calculated from input diagnostics, so we do not label this the "primary cause." Replacing only the Pc empirical formula or only RMW each fails to explain ~30%. The statement that "switching to a wind-field model that directly uses vmax (H10) would not rescue it either" is an **untested inference**: H10 `vmax_from_cen=False` was only checked in source code and has not been validated against observations. Restart condition: When a new AI synthetic catalog (or a new version of RAFT) becomes available, first screen it against the "near-station Cat3+ passage rate vs. HURDAT2" table (0 SU); only proceed into the design-wind-speed chain if all segments match the observational order of magnitude, re-preregistering under the original frozen thresholds and 19 stations; once the PI's wind-field model is ready, a separate trial of vmax-driven wind fields + RAFT can be conducted (validating the wind field against observations first). The local copy of the raw public data has been deleted from the workstation (it can be re-downloaded). - **10-14 23:30 Diagnostics v1 returned**: Replacing only the Pc empirical formula or only RMW each fails to explain ~30% (no separation experiment performed; we do not label a "primary cause"), leaving no pre-registerable fix, so no restart PREREG will be drafted; the remaining discrepancy lies in "intensity × passage frequency of strong near-station storms," and ideas has been dispatched for the final step: near-station strong-storm passage frequency in RAFT/STORM vs. HURDAT2 (deliverable), after which B2 will be wrapped up. - **10-14 22:45 Pre-restart diagnostic dispatched** (lab lead; status remains On hold): to ideas (queued), comparing vmax/Pc/ΔP/Rmax of RAFT/STORM against HURDAT2+EBTRK, swapping Pc and Rmax separately in the same H08 chain to inspect single-storm station peak wind changes, and checking CLIMADA wind field models that directly use vmax; without touching ASCE or recomputing return periods. Upon return, decide whether to draft a restart PREREG (report the draft to Node B, and report to the lab lead before running the gates). - Report errata (incorrect source for 14.5 storms/yr, relative comparison is not a gate, two undisclosed verification checks).

Opened 10-14 19:20; both EVA and Design Loads tracks confirmed no overlap and no borrowable station values or catalogs, and criteria and stations were frozen at 19:25 (including relative comparison rules if ASCE values are unavailable). Round 1 dispatched to ideas (data fetching, PREREG, dual-chain calculation, ASCE

value retrieval, GitHub repository table). Literature: Grok quick check available (clues only, unreviewed); Deep Research monthly quota exhausted; step 2 will follow once the quota resets, and litwiki will build cards from conclusions based on it.

Settled decisions

- 2026-10-14 22:30 Put on hold (review 6/6 in favor)
- 2026-10-14 19:25 Both tracks replied, criteria frozen (added relative comparison rules if ASCE values cannot be retrieved, plus read-only and prohibited items for Design Loads track materials) → "Supplement 2"
- 2026-10-14 Parent project added two boundaries: wait for replies from EVA / Design Loads tracks before freezing criteria; add stratified reporting by RMW distance and coastal distance → "Supplement"
- 2026-10-14 B2 kickoff, draft criteria and stations, Gate 0 prerequisite
- 2026-10-14 B2 opened (Node B lead)

Boundaries

- **Workstation does not run computations or store compute data (tightened by the PI on 10-14, recorded 10-15)**: All wind field / catalog diagnostics, batch runs, and large downloads must be executed and stored on Colab (CPU/GPU) or CRC/TACC; the workstation only holds scripts, small results <200 MB per item, figures, and reports; if temporary storage >1 GB is strictly necessary, report to Node B first. Public raw data (large RAFT/STORM/HURDAT2 files, GFS/ERA5 fields) belong in Colab sessions or TACC work storage. Supercomputer usage follows ComputeOps resource request + registration, with pre-check and submission separated into two commands; CRC defaults to a single node and the group storage is ~95% full, so large datasets go to TACC first. Task briefs upon any restart must follow this rule; "workstation local priority" in older entries below is void.
- No single-experiment cap on Colab, but pause and report to the Node B lead if projected usage is >100 CU; run `colab stop` when finished; dispatch only one executor per round. Do not download RAFT rainfall files, only the track `.nc` files; do not store >5 GB on the workstation.
- Do not use supercomputers. Check literature before making assumptions; the parameter source table is below. New data and code go in the project area; annotate data sources on all figures and tables.
- **EVA track reply (10-14, forwarded via parent project)**: EVA covers only thunderstorm high winds and mechanism-stratified extreme value analysis, with no overlap or restricted zones relative to B2; it has no ASCE 7-22 station values or hurricane catalogs to lend (ASCE materials belong to the Design Loads track, reply pending). Two caveats: ① B2's hurricane gust factors / averaging-time conversions must not be directly applied to thunderstorm or mixed-climate EVA (EVA candidate entry); ② if B2 results are later combined with the thunderstorm side into mixed-climate design winds, notify the EVA track via the parent project first to align conventions, and final conclusions on mixed-climate synthesis and return periods rest with the PI. Available for reference but likely unneeded: Cook ASOS four-station

Texas gust event inventory (read-only; not stratified by tropical cyclones, gusts not standardized to 3 s), not used in B2.

- Tier-2 node: do not open Tier-3 nodes; tool bots must not delegate further. EVA / Design Loads tracks are read-only; if their materials are needed, report to the Node B lead, who will request them.

What was tried and results

Date	What was done	One-line result
10-14 23:54	Passage-frequency attribution (observations only; + ADDENDUM)	Cat3+ passages within 100 km of station (per 1,000 storms): TX segment RAFT 0.09 / STORM 1.04 / Observed 3.45; LA–Panhandle 0.17 / 1.45 / 4.14; Florida Peninsula 1.97 / 2.55 / 2.07; Carolinas 0.42 / 2.66 / 3.45. RAFT does not lack near-station TS-level passages, but very few reach Cat3+ during passage (~20× lower in the Gulf of Mexico), whereas the Florida Peninsula is close to observations; this matches the segment-by-segment pattern of the V700 negative bias (deepest in the Gulf). Answer: relative to observations, it is (iii) both lower passage frequency and lower passage intensity (within 100 km, TS+ 0.168 vs. 0.321 and Cat3+[local path omitted]+ 6.2% vs. 11.6% for 1988–2018 observations; 0.315 and 12.1% for 1979–2018), whereas relative to STORM the difference lies mainly in passage intensity (6.2% vs. 16.0%); track geometry and overall occurrence rate cannot be separated, so this is correlative evidence only; "per 1,000 storms" in the table refers to per 1,000 qualifying basin-wide storms; the observed Cat3+ sample is small with wide segment-level annual-block intervals (TX [0.000, 0.086]), so only orders of magnitude are compared

Date	What was done	One-line result
10-14 23:20	Pre-restart diagnostic v1 (comparing observational inputs only, without touching ASCE; paired replacement on a 5% sample (2,351 storms) in the same H08 chain, baseline verified item-by-item against the main chain)	Near-station Cat3+ ΔP p90 in RAFT is 71.6 hPa (obs 87, STORM 88.9), and RMW distribution is narrow (p99 52 vs. obs 150 nmi). **Replacing Pc only**: median change in affected station peaks is 0.0%, Cat4/5 +1.84%, refuting reviewer E's "10–15% loss" (observational static eyewall check median 2.87%); **Replacing RMW only**: median +4.2%, whereas applying the same replacement to STORM yields +16.7% and node-by-node sampling introduces jumps, precluding a fixed correction. Neither explains the ~30% gap. CLIMADA 6.1.0 H10 `vmax_from_cen=False` allows catalog vmax to enter the wind field (source lines 249–318, 503–508), not yet validated against observations. Currently no pre-registerable correction exists
10-14 21:41	Integrated page numbers into new litwiki cards (PREREG_v7 source addendum, post-results, no parameter changes or re-judging)	Correction: "STORM 14.5 storms/yr" written in v1 has no basis in the original paper; the STORM generation paper gives North Atlantic Poisson λ=10.8 (KHFQRECZ PDF p. 4); STORM is always evaluated using its own 10,000-year statistics, so gate verdicts are unaffected; 14.5 is retained strictly as a RAFT annual-rate sensitivity value. RAFT authors' self-reported low bias is used only as a directional clue, not quantitative attribution
10-14 21:50	Switched annual rate to HURDAT2 contemporaneous, matched-threshold observational proxy (1979–2018, TS/HU ≥35 kt and ≥24 h, 483 storms = 12.075 storms/yr → 3,313 equivalent years; PREREG_v5/v6, post-processing recomputation only, 0 CU)	Primary gating version: Gate 0 passed (−8.87%, unchanged); Gate 1 still failed (RAFT 30.95%, >15% and >1.1×9.21%); sensitivity rates of 10/14.5/20 storms/yr and the run without the lifetime filter all yield 29–31%, leaving the verdict unchanged. Note: v6 was written to disk after the 14.5-version results (though the decision preceded the results), which ideas transparently documented

Date	What was done	One-line result
10-14 21:30	Round 1 v3 main chain: RAFT 40,000 tracks / STORM v4 10,000 years through the same H08 chain, 19 stations vs. ASCE 7-22 (public GIS, V700 19/19, V50 18/19 with KEYW missing)	Gate 0 passed (STORM V700 median −8.87%); Gate 1 failed (RAFT median absolute bias 30.40% > 15%, and > 1.1×STORM 9.21%); RAFT biased low across all four segments by −26% to −40%, not a single-station artifact; conversions 1.03–1.12, POCI ±5, and annual rates 10–20 do not change the verdict; RAFT V700 tail is supported by only ~3 storms; SPIN third chain (borrowing RAFT Pc/RMW formulas) V700 46.5–64.8 for reporting only; ~8.7 CU. Proposed to put on hold, pending review (6 reviewers)
10-14	Grok quick check (issued by Node B)	Existing design-wind studies all use hybrid methods; free-running AI tropical cyclone intensities are generally too weak; RAFT data are public (Zenodo 10392725, CC BY 4.0); 8/8 DOIs exist. Clues only

Parameter source table

(Wind field chain parameters carry over from the B1+B4 page parameter source table; this page lists only parameters newly added for B2) | Parameter | Value | Source or "assumption" | Places checked | | --- | --- | --- | --- | | STORM simulation years | 10,000 years (10 files of 1,000 years per basin; North Atlantic Poisson λ=10.8/yr, STORM generation paper PDF p. 4; "14.5/yr" was unfounded and has been corrected) | Bloemendaal et al. 2020 wind speed probability paper PDF p. 8 | Pending ideas verification of dataset documentation | | STORM wind averaging time | 10 m 10 min mean sustained wind | Same paper, PDF pp. 2, 8 | 10 min→1 min conversion factor to be checked (check literature before assuming) | | RAFT equivalent years / annual rate | 12.075 storms/yr (3,312.6 equivalent years), sensitivity 10/14.5/20 | Assumption (observational proxy): HURDAT2 2026-09-23 release, 1979–2018 TS/HU ≥35 kt and tropical lifetime ≥24 h, matching the 33 kt genesis threshold and <1 day exclusion in Xu 2024 PDF pp. 3–4 (PREREG_v6) | Xu 2024 PDF pp. 3–5, 9; Zenodo fields | | RAFT wind averaging time, RMW source, environmental pressure | To be checked | — | Same as above | | STORM/RAFT environmental pressure POCI | 1010 hPa, sensitivity 1005/1015 | Assumption (track files in both catalogs lack this field, which H08 requires) | Both original papers; Zenodo/4TU fields | | Output convention | 10 m, z0=0.03 m, 3 s gust | Vickery et al. 2010 PDF p. 8 | Exact ASCE 7-22 original text page numbers pending | | 1 min→3 s gust factor (open terrain) | Over-water 1 min → open-terrain 3 s combined multiplier 1.07, sensitivity 1.03–1.12 (PREREG_v1) | Primary candidate: Simiu et al. card (10 m open water 1 min ↔ open terrain 3 s, PDF pp. 1–3, Table 1; EVA-STEP3-C07 restricts use to matching conventions) | ASCE 7-22 commentary Durst curve; gust factor card | | Over-water baseline → open-land conversion | Retains v1 formula (500 m reference logarithmic profile, land z0=0.03 m) | B1+B4 page (assumption, range 200–1000 m) | Same as B1+B4 | | ASCE 7-22 station values | To be retrieved | Public sources (ATC Hazards by Location, etc.), without using the PI's credentials | — |

Open items and next steps

- **21:50 Annual-rate version returned**: Using the HURDAT2 observational proxy rate (12.075 storms/yr) as the primary gating version, RAFT is at 30.95% and still fails the gate, leaving the conclusion unchanged; the evidence package under review reflects the 14.5 version, and v6 will be verified together during the verdict.
- **21:30 Round 1 results returned**: Gate 0 passed, Gate 1 failed, proposed to put on hold; per protocol, abandoning a route requires review first; the review (6 reviewers) is running, with status to be finalized after the verdict. The annual rate still used 14.5/yr (not yet updated to HURDAT2 as requested at 19:54), but the 10–20/yr sensitivity shifts V700 by only −3.4%/+1.5%, which cannot close a 30% gap; submitted to reviewers for confirmation.
- 20:35 check: Supplement reported back: Supplements 1 and 2 were already incorporated into the existing plan, so no duplicate runs were launched; **ASCE 7-22 V700 values have been retrieved from public GIS for all 19 stations** (≥12 stations, so Gate 0 / Gate 1 are judged as usual without falling back to relative comparison). Before starting the main run, ideas caught an RMW unit bug during self-check (CLIMADA `tctrack_to_si` expects nautical miles, whereas the original script passed meters, causing double conversion), wrote PREREG_v3 (20:06) voiding the first 6 batches, and reran; simultaneously changed the handling of Pc>POCI to follow CLIMADA's default clipping (v1 originally stated "discard"). All changes occurred before any station statistics or gate evaluations, with zero parameter tuning. Separately wrote PREREG_v4_SPIN (third chain for reporting only, borrowing RAFT's empirical Pc/RMW formulas and assuming POCI = 1010). The B2 lead re-verified the B1+B4 chain: B-deck RMW is natively in nautical miles and passed directly, so **B1+B4 results are unaffected by this bug**.
- Note: The file named PREREG_v3 by ideas is the unit fix, not the change of RAFT annual rate to HURDAT2 requested by the B2 lead at 19:54 (still queued, to become the next PREREG version); that decision preceded any results, and because annual rate only scales empirical return periods, it can be updated in post-processing, with the HURDAT2 annual-rate version serving as the authoritative gating version.
- 19:54 check: ideas has started work; PREREG_v1 (19:50) and v2 (switching STORM to the deduplicated v4 release) have been written, and no station wind speeds have been computed yet. The B2 lead has instructed that v3 be issued prior to computation: set the primary RAFT annual occurrence rate to the contemporaneous, matched-threshold HURDAT2 observed annual mean, and downgrade STORM's 14.5 storms/yr to a sensitivity case.
- (Completed) Round 1 and annual-rate version; now on hold
- (Completed 10-14) 7 litwiki cards created; Bose 2023 final publication page numbers not yet verified, so its page numbers are not cited for now
- Deep Research step 2: send once the quota resets; conclusions to follow

Literature basis

- 2026-10-16 data: 7,524 NOAA HRD dropsonde 10–300 m tropical cyclone wind profiles (Ma et al. 2026, Zenodo 10.5281/zenodo.21868265, CC BY 4.0; approved by the PI with instructions to archive; backed

up on CRC). Useful for profile shape, 10–300 m wind shear, comparison against H08/empirical profiles (over ocean), and profile constraints if B1+B4 is restarted; not overland profiles. Does not alter this node's status or gates.

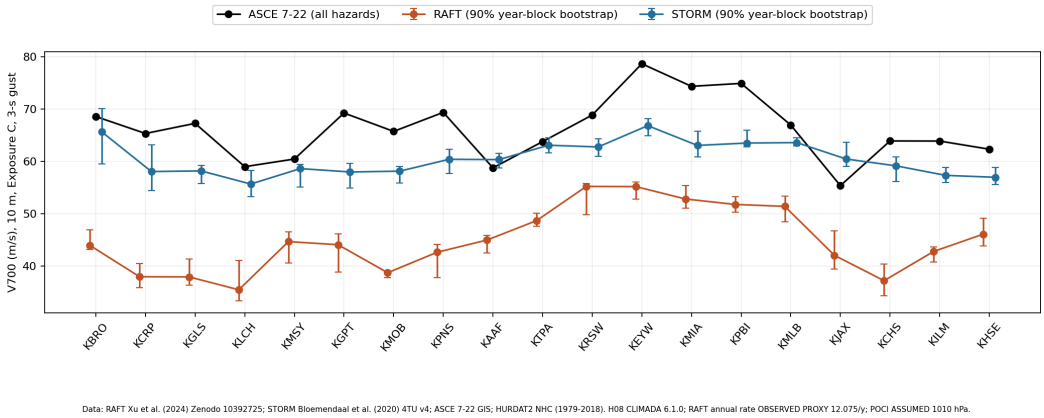
- Cards (created by litwiki on 10-14; see the index):
- RAFT: the `year` field denotes the environmental year (PDF pp. 3, 5); intensity is 10 m 1 min, Pc and Rmax are given by empirical formulas, and no environmental pressure is provided (PDF p. 4 Eqs. 2–3); the authors self-report that across 51 US coastal cities, landfall 99th-percentile wind is ~8 kt low and Rmax is ~19% low on average (PDF pp. 6–7). This points in the expected direction of a low bias for the RAFT chain.
- STORM generation paper; wind speed probability paper.
- Vickery 2010: 10 m open-terrain peak gust, Category II = 700 years.
- SPIN: full synthetic track set is publicly available, Zenodo 10.5281/zenodo.20795548, CC BY 4.0 (PDF p. 19 / journal p. 7321) → usable as a third chain for reporting only.
- Bose 2023: catalog available only "upon reasonable request," not public; final PDF page numbers unverified.
- Snaiki & Wu 2019.
- Grok quick-check report (clues, unreviewed): see table above
- ASCE 7-22:
- Vickery 2000 track model:

Notes from the PI

10-22 citation snowballing

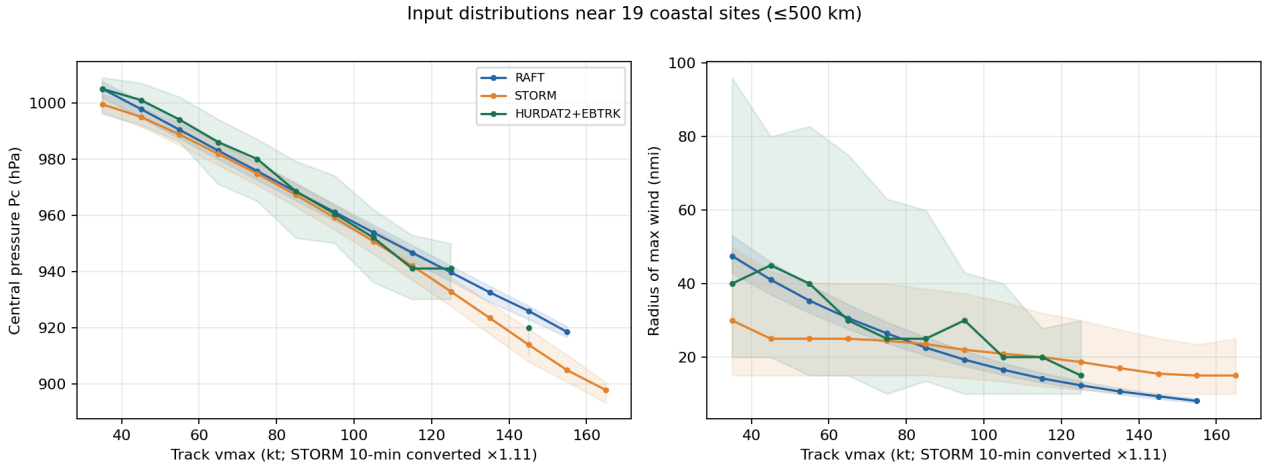
- Citation pool (initial run truncated at 5,000, rerun with a 12,000 cap). Selected 9 papers for LitWiki Batch 8, focusing on: regional loss discrepancies across global synthetic tropical cyclone models (s41467-022-33918-1), impact of wind-field model choice on typhoon risk (jweia 2021.104792), and increasing US coastal hurricane risk (sciadv.adf0259). At the title level, these align with "~31% negative bias under the catalog + H08 configuration and too few strong nearshore hurricanes," leaving the stop category unchanged; magnitudes will be checked once the cards are built.
- 10-22 Batch 8 verification (#8–16): None replicated or refuted the ~31% negative bias in V700 under RAFT + H08; multiple papers support that "synthetic catalog and climate assumptions strongly affect return-period values and losses" (Meiler 2022: 100-year North Indian Ocean losses range from 40 to 246 billion USD across four catalogs, the lead spot-checked PDF p.1 and p.3) and that "nearshore frequency and intensity must be decomposed separately" (#9, #12). Note that #16 Huang 2021 compares intensity models rather than wind field models and cannot be used as magnitude evidence for wind-field model selection. Stop category unchanged.

Key figures



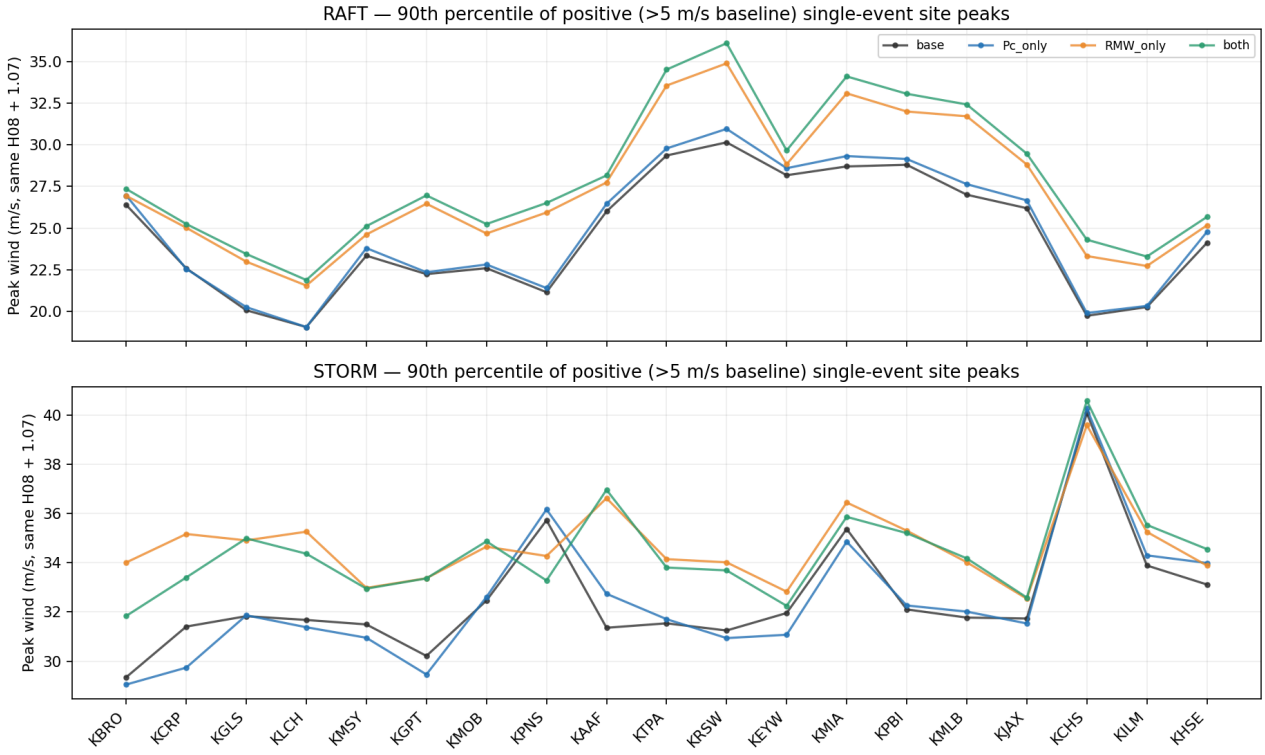
Data: RAFT Xu et al. (2024) Zenodo 10392725; STORM Bloemendaal et al. (2020) 4TU v4; ASCE 7-22 GIS; HURDAT2 NHC (1979-2018). H08 CLIMADA 6.1.0; RAFT annual rate OBSERVED PROXY 12.075/y; POCI ASSUMED 1010 hPa.

v700 comparison



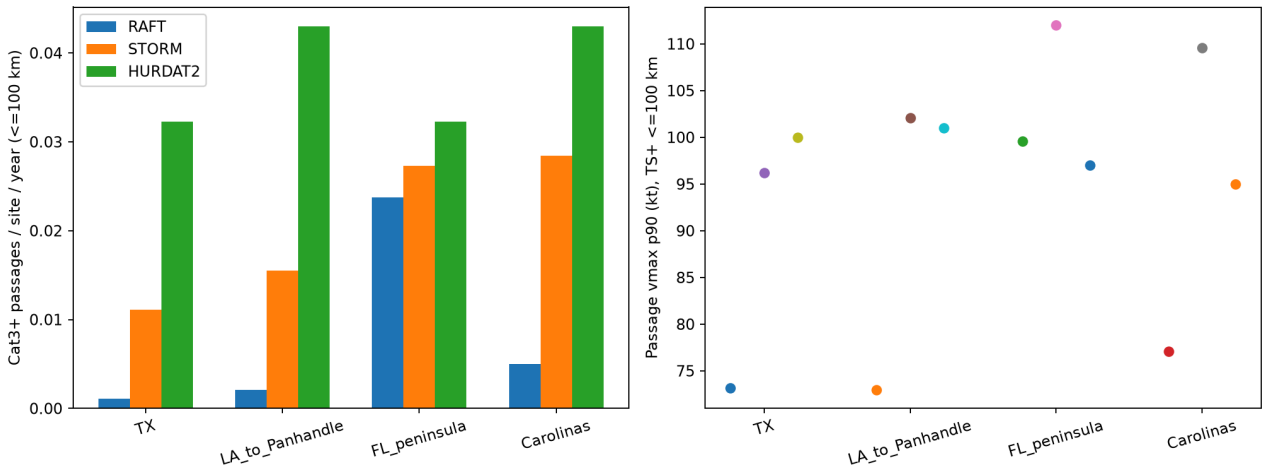
Sources: Xu et al. 2024 RAFT; Bloemendaal et al. 2020 STORM v4; NHC HURDAT2 2026 + CIRA EBTRK v3.0.2. Native track nodes; shading P10–P90.

inputs comparison



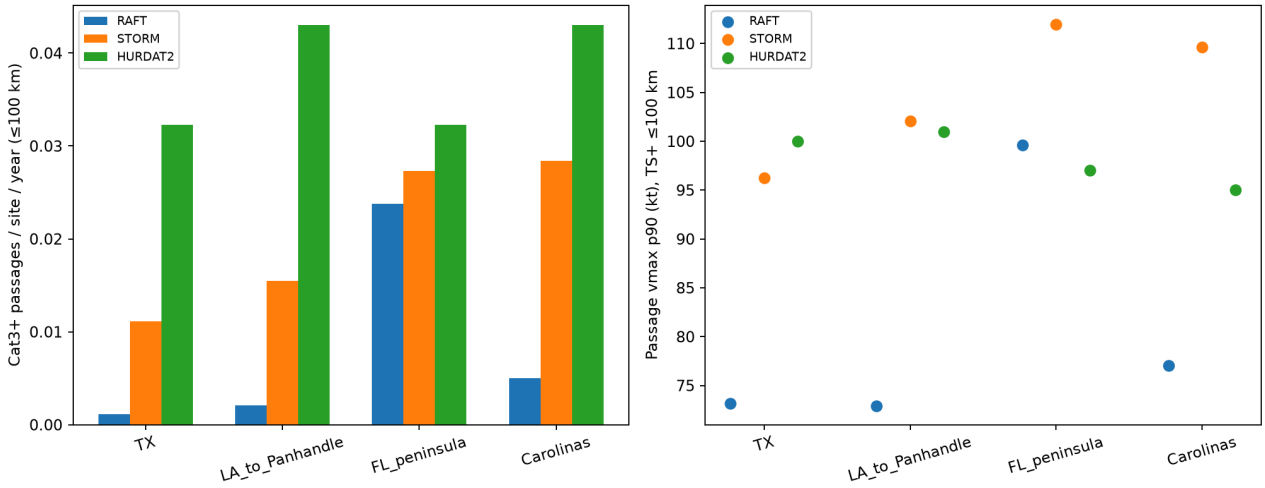
Sources: RAFT (Xu 2024), STORM v4 (Bloemendaal 2020), HURDAT2 + EBTRK; CLIMADA 6.1.0 H08. Deterministic 5% event sample; NOT return levels.

paired peak 5pct



Sources: RAFT Xu 2024; STORM v4 4TU; HURDAT2 NHC 2026; EBTRK CIRA. Native nodes, no hazard-model winds.

passage v1



Sources: RAFT Xu 2024; STORM v4 4TU; HURDAT2 NHC 2026; EBTRK CIRA. Native nodes, not model winds.

passage v1 legend